

AI-Based Fake Identity & Document Screening System: Pipeline & Architecture

This document outlines the comprehensive architectural blueprint for the AI-powered identity screening platform designed for border checkpoints [cite: 1]. The system augments human judgment by reducing document verification time from minutes to seconds, providing explainable risk scores, and generating immutable digital audit trails [cite: 1].

1. High-Level Architecture & Pipeline Flow

The pipeline is built on an event-driven microservices pattern connected via Apache Kafka [cite: 1]. This decouples heavy GPU operations from CPU tasks and ensures scalability under peak passenger loads [cite: 1]. The flow is executed as follows:

1. **Capture & Preprocessing:** The gateway ingests the document, applies deskewing and denoising, and classifies the document type [cite: 1].
2. **Kafka Fan-Out:** The *doc.preprocessed* topic triggers three parallel microservices simultaneously [cite: 1].
3. **Parallel AI Processing:** OCR Extraction, Tampering Detection, and Face Verification execute concurrently on the image payload [cite: 1].
4. **Rule Validation:** Text outputs from the OCR pipeline are validated against formatting regex and watchlist databases [cite: 1].
5. **Risk Fusion Engine:** A windowed join aggregates all outputs using the *doc_id* partition key [cite: 1]. The engine computes a final risk score [cite: 1].
6. **Decision Dashboard & Audit:** A human officer makes the final adjudication based on explainable factors, and the decision is hashed to a permissioned blockchain [cite: 1].

2. Microservices, Architectures, and Pretrained Models

The system utilizes specialized architectures tailored to the unique physical properties of

document anomalies, avoiding one-size-fits-all models [cite: 1].

Module	Core Architecture / Model	Technical Justification
Preprocessing	MobileNetV3-Small	Classifies document types to map spatial field templates [cite: 1]. ImageNet pretraining is effective here for natural-image semantic content [cite: 1].
OCR Engine	DBNet/CRAFT, PaddleOCR, LayoutLMv3	Segmentation-based text detectors handle dense, rotated text better than box regression [cite: 1]. LayoutLMv3 fuses text and spatial position for semantic mapping [cite: 1].
Tampering Detection	Custom Shallow CNN (~1.5M params)	Analyzes physically-grounded weak signals (ELA heatmaps, PRNU noise residuals) [cite: 1]. Pretrained ImageNet models are avoided because they suppress the high-frequency noise required for this task [cite: 1].
Face Verification	Silent-Face-Anti-Spoofing & ArcFace	A dedicated liveness gate prevents presentation attacks [cite: 1]. ArcFace (ResNet-50) is used for high-confidence identity embedding [cite: 1].

3. Risk Fusion Engine & Decision Scoring

The Risk Fusion Engine is the aggregator located immediately before the officer's dashboard. It merges the parallel signal streams into an explainable 0-100 continuous score [cite: 1].

- **Hard Overrides:** Near-certain fraud signals (e.g., a watchlist hit or liveness failure) bypass regression and force a floor score of 95 or 92 [cite: 1].
- **Regression Engine:** For documents without hard flags, a Gradient Boosted Tree (e.g., XGBoost) regressor processes the signals [cite: 1]. Monotonic constraints ensure legally defensible scoring (e.g., higher tamper signals strictly result in higher fake scores) [cite: 1].
- **Explainability:** SHAP (Shapley Additive exPlanations) values are extracted directly from the tree ensemble to provide the officer with the exact top-3 contributing factors [cite: 1].

4. Proposed Database Schema (State Store)

A relational database stores the operational state of documents moving through the Kafka topics [cite: 1].

Table: documents

- doc_id (UUID, PK): Partition key matching the Kafka topics.
- status (VARCHAR): Tracks progress (ingested, processing, scored).
- created_at (TIMESTAMP)

Table: ocr_extractions

- doc_id (UUID, FK)
- mrz_valid (BOOLEAN): ICAO 9303 checksum validation result.
- extracted_json (JSONB): Mapped semantic fields and confidence levels.

Table: tamper_analysis

- doc_id (UUID, FK)
- tamper_score (FLOAT): Raw output from the custom fusion CNN.
- flagged_regions (JSONB): Bounding boxes for ELA/PRNU anomalies.

Table: face_verification

- doc_id (UUID, FK)
- liveness_passed (BOOLEAN): Hard sequential gate output.
- similarity_score (FLOAT): Cosine similarity calculation.

Table: risk_assessments

- doc_id (UUID, FK)
- final_score (FLOAT): Probability output from XGBoost (0-100).
- top_factors (JSONB): SHAP values for the top 3 risk factors.
- officer_override (INTEGER): Adjudication adjustment (-10, +10, +20).
- audit_hash (VARCHAR): Hashed payload committed to Hyperledger Fabric.

5. Generated Documents & Artifacts

Throughout the training and execution phases, specific data artifacts are generated to support the AI pipeline [cite: 1]:

- **Synthetic Forgery Images:** Generated programmatically by pasting donor patches (faces/text) onto base IDs and saving at varying JPEG compressions to lock in compression history artifacts [cite: 1].
- **Forensic Feature Tensors:** Extracted ELA heatmaps and PRNU residual arrays, precomputed to serve as the direct input channels for the custom Tampering CNN [cite: 1].
- **Simulated Tabular Datasets:** Monte Carlo simulations generating 9-dimensional numerical feature vectors to bootstrap the XGBoost regressor before real operational data is accumulated [cite: 1].
- **Audit Ledger Hashes:** Cryptographic representations of the complete verification and officer decision payload, stored immutably on a permissioned Hyperledger Fabric blockchain [cite: 1].